

Project Synopsis

TruLabel

An Agentic AI System for Verifying Product Marketing Claims Against Credible Evidence

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1. Project Title

TruLabel: An Agentic AI System for Verifying Product Marketing Claims Against Credible Evidence.

2. Objectives

- To design and build a multi-agent AI system that automatically extracts checkable marketing claims from a product's advertised description or label.
- To verify each extracted claim against credible, independently sourced ingredient/nutrition data and regulatory reference material.
- To produce a transparent, evidence-cited Marketing Accuracy Score (MAS) that helps consumers make informed purchase decisions.
- To extend single-product verification into comparison and recommendation capabilities across multiple products in the same category.

3. SDG Alignment

SDG	Alignment
SDG 12 — Responsible Consumption and Production	Primary alignment. TruLabel directly targets misleading marketing that drives uninformed consumption, helping consumers verify claims before purchase and encouraging producers toward more accurate labeling.
SDG 3 — Good Health and Well-being	Secondary alignment. Verifying health/wellness claims on food products (e.g., "no added sugar," "immunity boosting") helps prevent consumers from being misled about the health impact of what they consume.

4. Existing System and Its Limitations

Consumers currently rely on a combination of the product's own marketing copy, generic online reviews, and crowd-sourced ingredient databases (e.g., Open Food Facts, viewed manually) to judge a product's authenticity. Existing tools have significant gaps:

- Generic review platforms assess overall satisfaction, not the truthfulness of specific marketing claims.
- Ingredient/nutrition databases (like Open Food Facts) present raw data but do not cross-reference it against what a product's marketing actually claims.
- No consumer-facing tool in the Indian market systematically checks a product's claims against regulatory definitions (e.g., FSSAI's Advertising & Claims Regulations) or documented misleading-advertisement rulings (CCPA).
- Comparing multiple products' marketing accuracy, or discovering a genuinely better alternative, currently requires the consumer to do all the cross-referencing manually.

5. Proposed System

TruLabel proposes an agentic pipeline that automates this verification process end to end. A user searches for or submits a product; the system extracts its marketing claims, retrieves real ingredient/nutrition ground truth, checks each claim against that data and against curated regulatory reference material, weighs the credibility of each evidence source, and produces a plain-

language verdict with an overall Marketing Accuracy Score. The system further supports comparing multiple products side by side, and recommending a better-scoring alternative within the same product category.

6. Expected Outcomes

- A working, deployed multi-agent system capable of verifying real product claims against real evidence sources.
- A validated Marketing Accuracy Score methodology, tested against a curated set of known misleading-marketing cases (CCPA rulings).
- A comparison and recommendation capability demonstrating agentic decision-making beyond single-query response generation.
- A full-stack, demoable web application suitable for a final evaluation and viva.

7. Specialization Concepts Used

- Agentic AI / multi-agent orchestration (sequential agent pipeline with defined input/output contracts).
- Large Language Model (LLM) reasoning for claim extraction and evidence synthesis.
- Information retrieval and heuristic ranking (for category-based candidate filtering in Recommend Mode).
- Full-stack software engineering: REST API design, relational database design, frontend-backend integration.
- Applied data ethics: source credibility weighting and transparent, evidence-linked verdicts.

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